

## Unit 6 · A 5D origin of scale duality: a compactified spacetime scale fifth dimension

*What the five-dimensional embedding generates,  
and what it cannot predict*

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### ABSTRACT

**Context.** Units 1–3 treat the scale axes ( $\psi$  spatial,  $\chi$  temporal) as *internal targets* of a sigma model — explicitly *not* spacetime dimensions — so as to avoid ghosts and closed timelike curves (result F2).

**Problem.** We study the *opposite* choice: promoting the scale axis to a genuine fifth spacetime dimension. Two questions: does this embedding *regenerate* the corpus action? and does it allow *predicting* the coupling  $\beta$  rather than fitting it? **Method.** Kaluza–Klein reduction of a warped 5D metric along a compact scale circle, with symbolic verification of all the tensor algebra, spectral computation of the KK tower, and heat-kernel expansion of the deformed operator. **Results.** (i) *Generative* [proof] [numerical]: the KK reduction reproduces the corpus structure — induced non-minimal coupling  $f(\sigma) = a(\sigma)^2$ , trigonometric form  $f = 1 + \beta \cos \sigma$  forced by compactness,  $V$ – $f$  locking giving exactly  $V = \alpha(1 - \cos \sigma)$ , and the F2 health criterion *re-derived* (spacelike scale axis  $\rightarrow$  healthy; timelike  $\rightarrow$  ghost); for  $a = e^{\sigma/\ell}$  the geometry is exactly  $\text{AdS}_5$ , the scale playing the role of the holographic / renormalization-group direction,  $\partial_\sigma$  its dilatation isometry and  $\sigma \rightarrow -\sigma$  a scale mirror (cousin of T-duality). The discreteness of the KK tower further provides a *bridge* (candidate, not derived) toward the  $\tau_0, \varepsilon$  clock of Unit 1. (ii) *Cost* [proof]: the promotion *reopens* the ghost / closed-timelike-curve question that F2 closed by postulate (closed here by the choice of spatial signature), and closing the warp into a loop requires a violation of the null energy condition (NEC) in the bulk, localized at the Planck pole  $P$ . (iii) *Conjecture* [conjecture]: this exotic source is identified with vacuum Casimir energy — correct tensor structure  $\text{diag}(1, 1, 1, 1, -4)$ , correct sign (violating the NEC) for periodic fields, correct Planckian magnitude (self-consistent once  $\ell \sim \ell_5$ ). (iv) *Rigorous negative* [proof]: vacuum energy does *not* predict  $\beta$ . The third heat-kernel coefficient of the warped operator is non-zero,  $b_3(\beta) \simeq 0.146\beta^2$ , so the 5D Casimir energy is *ultraviolet-divergent* at the very order that would fix  $\beta$ ; the value  $\beta_*$  depends on the renormalization scheme. The corpus  $\sigma$ -model formulation does not have this bulk divergence — a point in its favor. **Conclusion.** The 5D embedding is a *structural origin* (generative), not a more predictive replacement;  $\beta$  remains a coupling constrained by the CMB, as in the corpus.

**Keywords:** scale duality; Kaluza–Klein reduction; warped extra dimension;  $\text{AdS}_5$  / renormalization direction; induced non-minimal coupling; Casimir energy; null energy condition; Seeley–DeWitt coefficients; natural inflation.

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**Epistemic status:** [proof] exact algebra / symbolic computation · [numerical] reproducible integration · [conjecture] plausible, not proven · [heuristic] analogy · [speculation] open · [to check] external ref. · [open] to be established. No conjecture is presented as a theorem.

**Preliminary note.** This unit explores a geometric choice deliberately opposite to that of the corpus, for the sole purpose of testing whether the scale-duality action has a dimensional origin. The verdict is mixed and we record it as such: the embedding is remarkably generative, but it *does not improve* the predictive power of the framework and, on one precise point (the ultraviolet behavior of the vacuum), it is less well-behaved than the original  $\sigma$ -model formulation. The formalization was carried out in dialogue with analytical-assistance systems, under the direction of the author, who bears full responsibility for the scientific content.

## 1 Introduction: how and why a fifth scale dimension

### 1.1 The corpus choice, and its opposite

The corpus (Units 1–3) rests on one hypothesis: the logarithmic scale axis  $u = \ln(\lambda/\ell_P)$ , usually non-compact, is in fact a circle, via the projection  $\psi = 2 \arctan u \in (-\pi, \pi]$ , which identifies the Planck pole ( $\psi = 0$ , denoted  $P$ ) and a “unified” pole ( $\psi = \pm\pi$  identified, denoted  $\infty^*$ ). One point is crucial there: these scale circles are *internal targets* of a Euclidean sigma model, *not* spacetime dimensions. This is result F2: since  $\chi$  is not a timelike coordinate, there is neither ghost nor closed timelike curve. This choice is *conservative*: it imports the duality structure without touching the spacetime signature.

The question of this unit is the *opposite* choice. *What if* one took the scale axis literally, as a genuine fifth spacetime dimension? This choice is bolder: it reopens the ghost and CTC question that F2 closed by postulate. But it offers a possible payoff: if the corpus action *descends* from a 5D geometry by dimensional reduction, then its particular form (the locked non-minimal coupling, the bounded potential) ceases to be a choice and becomes a *consequence*. This is the generative stake we test.

### 1.2 Why “spacetime”: the relativistic locking

The corpus has *two* scale axes: a spatial axis  $\psi$  (length scale  $\lambda$ ) and a temporal axis  $\chi$  (duration scale  $\tau$ ). A fifth dimension is *unique*. For a single dimension to carry both the spatial scale and the temporal scale, these two scales must be *locked*: this is the relativistic locking  $\tau = \lambda/c$ , i.e. a dynamical exponent  $z = 1$ . Under this locking, the two axes  $\psi$  and  $\chi$  collapse onto a single diagonal coordinate, which we denote  $\sigma$ . The fifth dimension is thus *spacetime* in the precise sense that it simultaneously encodes both scales, tied by  $c$  — not in the sense of a mixed signature.

**Remark 1 — dictionary with Units 2–3** In this unit,  $\beta$  denotes the amplitude of the 5D non-minimal coupling along the diagonal axis  $\sigma$ . Under the identification  $\sigma \leftrightarrow (\psi = \chi)$ , it coincides with the *diagonal* non-minimal coupling of the corpus, i.e. twice the per-axis coupling. The diagonal closure threshold  $\beta_c^{\text{diag}} = -1/3$  thus corresponds to the per-axis threshold  $-1/6$  of Units 2–3, and the viable band translates accordingly. All the formulas below are written with the diagonal  $\sigma$  and stay within the “single  $\beta$ ” convention.

### 1.3 One-point compactification and duality involution

The compactified scale axis identifies the infinitely small and the infinitely large. The projection  $\sigma = 2 \arctan u$  sends  $u = 0$  (Planck) to  $\sigma = 0$  ( $P$ , true vacuum) and  $u \rightarrow \pm\infty$  (extreme scales) to  $\sigma = \pm\pi$  ( $\infty^*$ , false vacuum), the two poles  $\pm\pi$  being identified. The scale duality  $\lambda \mapsto \ell_P^2/\lambda$  is the involution  $D: \sigma \mapsto -\sigma$ , whose  $P$  and  $\infty^*$  are the two fixed points. It is this  $\mathbb{Z}_2$  involution that will constrain the form of the warp (§4).

### 1.4 The geometric anchor: scale as a holographic direction

The choice is not gratuitous. A 5D metric warped along the scale coordinate,

$$ds_5^2 = a(\sigma)^2 \eta_{\mu\nu} dx^\mu dx^\nu + \ell^2 d\sigma^2, \quad \mu, \nu = 0, 1, 2, 3, \quad (1)$$

takes, for the exponential choice  $a(\sigma) = e^{\sigma/\ell}$ , the exact form of anti-de Sitter space  $\text{AdS}_5$  in Poincaré coordinates. Now, in the AdS/CFT correspondence, the radial coordinate of  $\text{AdS}_5$  is the energy scale of the boundary theory, i.e. the renormalization-group direction. Identifying the scale axis with a fifth dimension is therefore to rediscover the holographic intuition: “moving in scale” = “moving in a dimension”. The compactification into a circle (instead of the radial half-line of AdS) is what distinguishes our framework and breaks the exact dilatation, generating a hierarchy of scales between  $P$  and  $\infty^*$ . The program is then clear: *reduce* (1) over  $\sigma$  and check what the 4D inherits.

## 2 Formal framework: warped metric and Kaluza–Klein reduction

**Postulate 1 — 5D arena:** The fundamental spacetime is  $\mathcal{M}_5 = \mathcal{M}_{3+1} \times S_\sigma^1$ , where  $S_\sigma^1$  is the scale circle (period  $2\pi$ ),  $\sigma = 0 \equiv P$ ,  $\sigma = \pm\pi \equiv \infty^*$ . The metric is (1) with  $a(\sigma) > 0$  periodic. The coordinate  $\sigma$  is spacelike ( $g_{\sigma\sigma} = +\ell^2$ ), the only choice compatible with the absence of ghosts (§4, F2).

[proof] (verified by symbolic computation) For the metric (1), the non-zero components of the 5D Einstein tensor are, writing  $' = d/d\sigma$  and  $\ell = 1$ ,

$$G^t_t = G^x_x = 3\left(\frac{a''}{a} + \frac{a'^2}{a^2}\right), \quad G^\sigma_\sigma = 6\frac{a'^2}{a^2}, \quad R_5 = -8\frac{a''}{a} - 12\frac{a'^2}{a^2}. \quad (2)$$

The reduction of the 5D Einstein–Hilbert action  $\int d^5x \sqrt{-g_5} R_5$  over the circle produces, in the 4D Jordan frame, a tensor–scalar system whose non-minimal coupling is the *square of the warp*,

$$\boxed{f(\sigma) = a(\sigma)^2} \quad (\text{gravity induced by the warping}). \quad (3)$$

## 3 General geometric properties

Before the reduction, the metric (1) has a geometric structure that clarifies *why* the identification scale = dimension is natural — and what it costs.

### 3.1 AdS<sub>5</sub> and the dilatation isometry

[proof] In the current region, the exponential warp  $a(\sigma) = e^{\sigma/\ell}$  gives

$$ds_5^2 = e^{2\sigma/\ell} \eta_{\mu\nu} dx^\mu dx^\nu + \ell^2 d\sigma^2, \quad R_5 = -\frac{20}{\ell^2} \text{ (constant)}, \quad (4)$$

i.e. *exactly* anti-de Sitter space AdS<sub>5</sub> of radius  $\ell$ . The shift  $\sigma \rightarrow \sigma + \lambda_0 \ell$ ,  $x^\mu \rightarrow e^{-\lambda_0} x^\mu$  leaves  $ds_5^2$  invariant: this is the dilatation, whose Killing vector is  $\partial_\sigma$ . [to check] (standard reading) the scale coordinate  $\sigma$  is then the radial / holographic direction of AdS, identified with the renormalization scale; the Noether charge of  $\partial_\sigma$  is a genuine *scale momentum* — the dynamical analogue of the topological winding charge of Units 2–3. The intuition “scale = dimension” therefore lands on marked-out ground, not out of nowhere.

### 3.2 The duality involution: a scale mirror

[proof] The map  $\mathcal{D} : \sigma \rightarrow -\sigma$ , i.e.  $\lambda \rightarrow \ell_P^2/\lambda$ , exchanges ultraviolet and infrared — the analogue of string T-duality. Its fixed points are exactly  $\sigma = 0$  ( $P$ ) and  $\sigma = \pi$  ( $\infty^*$ ): a scale mirror that leaves the Planck point and the confluence point invariant. Keeping the temporal axis independent, one recovers the Klein group  $\mathbb{Z}_2 \times \mathbb{Z}_2$  of Units 2–3.

### 3.3 Compactness breaks dilatation — and that is an asset

[proof] A *periodic* warp cannot be the exact exponential (4) (which is not periodic): periodicity forces the bounded form  $a^2 = 1 + \beta \cos \sigma$  and therefore *breaks* the exact dilatation isometry. Far from being a defect, this breaking is precisely what *generates a hierarchy of scales* ( $\ell_P$  versus the cosmological scale) between  $P$  and  $\infty^*$ , instead of an exact scale invariance that nothing observes.

### 3.4 Kaluza–Klein tower and a candidate origin for the clock (bridge to Unit 1)

[conjecture] Since the extra dimension is a circle, every field there has a *discrete* spectrum (§6, Fig. 3): zero mode = 4D physics, massive modes quantized by  $1/\ell$ . This discreteness is the geometric candidate for the origin of the *intermittent clock* of Unit 1: the oscillation in the compact direction  $\sigma$  would play the role of the pendulum whose average gives the existence factor  $\varepsilon = E(k)/K(k)$  and the period  $\tau_0 = 2\pi\sqrt{\omega/\alpha}$ . The scale coherence is encouraging — the fundamental KK frequency  $\sim 1/\ell$  fixes a period  $\sim \ell$ , and  $\ell \sim \sqrt{\omega/\alpha}$  recovers the right order of magnitude for  $\tau_0$  — but this is *not* a derivation: we have not established it in this work. It is the direction that would turn a conjecture of Unit 1 into a result, and we record it as such, without overstating it. [open]

### 3.5 Degeneracy at the confluence point

[open] The exponential reading (4) would suggest a degeneracy at  $\infty^*$ , where the two limits of the warp ( $a \rightarrow \infty$  and  $a \rightarrow 0$ ) would be glued together. But the *periodic* warp regularizes this gluing: at  $\sigma = \pi$ ,  $a^2 = 1 - \beta$  stays *finite* (the 4D slice does not degenerate). The apparent singularity of the exponential is therefore an artifact of its global extrapolation; the full consistency of the stereographic compactification of scale with the warp dynamics, however, remains to be established.

## 4 The generative content

### 4.1 Trigonometric form forced by compactness

[proof] Since the coordinate  $\sigma$  is a compact angle,  $f(\sigma)$  must be  $2\pi$ -periodic; invariance under the duality involution  $D : \sigma \rightarrow -\sigma$  imposes parity. The lowest non-trivial harmonic form is therefore

$$f(\sigma) = 1 + \beta \cos \sigma, \quad a(\sigma) = \sqrt{1 + \beta \cos \sigma}, \quad (5)$$

with  $f(P) = 1 + \beta$  (true vacuum) and  $f(\infty^*) = 1 - \beta$  (false vacuum). Positivity  $f > 0$  requires  $|\beta| < 1$ . This is exactly the non-minimal structure of the corpus, here *forced* by the compact geometry rather than postulated.

### 4.2 $V$ - $f$ locking: the bounded potential descends from the warp

[proof] We seek the simplest 4D potential compatible with compactness: periodic, *affine in  $f$*  (hence generated by the same source as the coupling), and vanishing at the true vacuum,  $V(P) = 0$ . These three conditions determine it uniquely. Writing  $V = A + Bf$  and imposing  $V(P) = A + B(1 + \beta) = 0$ , one obtains

$$\boxed{V(\sigma) = \frac{\alpha}{\beta} (1 + \beta - f) = \alpha (1 - \cos \sigma)}, \quad V(P) = 0, \quad V(\infty^*) = 2\alpha, \quad (6)$$

i.e. *exactly* the bounded potential of the corpus. A brane-tension term, which would behave like  $a^4 = f^2 = (1 + \beta \cos \sigma)^2$ , would inject a second harmonic  $\cos 2\sigma$  (via  $\cos^2 \sigma$ ) absent from (6): the potential therefore does *not* descend from a brane tension, but from a source in  $a^2$ . The coupling and the potential thus share a single geometric origin.

### 4.3 The cost, and the F2 criterion re-derived from 5D

[proof] *The cost of the promotion.* Making scale a genuine dimension *reopens* the question that the corpus had closed by postulate (F2): a *timelike*, compact extra dimension generates closed timelike curves and ghosts. We close it again, but only at the price of a signature choice. The kinetic term of  $\sigma$  arising from the reduction carries the sign of  $g_{\sigma\sigma}$ . For spacelike  $\sigma$  ( $g_{\sigma\sigma} = +\ell^2$ ), the 4D kinetic term has a healthy sign; for timelike  $\sigma$ , it becomes a ghost (opposite sign, verified by the determinant of the induced metric). The corpus result F2 — absence of ghost and CTC — therefore appears here as a *consequence* of the signature choice of the fifth dimension, where the corpus posited it. This choice (spacelike) is that of Postulate 2.

### 4.4 The cost: null energy condition violation at the Planck pole

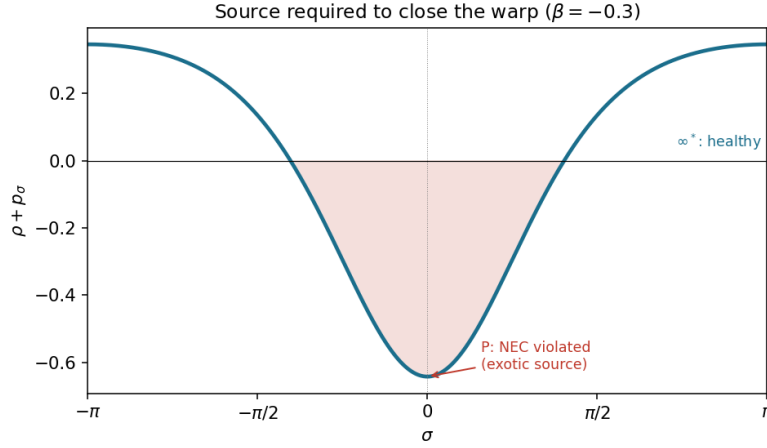
[proof] For a *periodic* warp (5) (which must come back down after rising, in order to close) to solve the 5D Einstein equations, the bulk source must violate the null energy condition along  $\sigma$ . The relevant combination is read off (2):

$$\kappa^2 (\rho + p_\sigma) = G^\sigma_\sigma - G^t_t = -3(\ln a)'' = \frac{3\beta(\beta + \cos \sigma)}{2(1 + \beta \cos \sigma)^2}. \quad (7)$$

For the physical sign  $\beta < 0$ , this quantity is *negative* (NEC violated) in the neighborhood of  $P$  ( $\sigma = 0$ :  $3\beta/[2(1 + \beta)] < 0$ , minimum of the warp) and positive (healthy) in the neighborhood of  $\infty^*$  (Fig. 1). The violation is therefore localized exactly at the Planck pole.

## 5 The exotic source: vacuum Casimir energy

The decisive question is the *nature* of this source. Is it an ad hoc defect (fatal), or a natural quantum effect? We show that it is Casimir energy.



**Figure 1.** [numerical] Required source  $\rho + p_\sigma$  along the scale circle ( $\beta = -0.3$ , Eq. (7)). The null energy condition is violated (red region,  $\rho + p_\sigma < 0$ ) in a neighborhood of the Planck pole  $P$ , where the warp reaches its minimum; it is healthy near  $\infty^*$ . Closing the scale line into a loop has a cost, and this cost concentrates at Planck.

### 5.1 Tensor structure

[proof] Decomposing the required source  $\kappa^2 T_B^A = G_B^A$  into a trace part (bulk cosmological constant) and a traceless part, the latter equals

$$\tau_B^A = \frac{3}{5} (\ln a)'' \text{diag}(1, 1, 1, 1, -4), \quad (8)$$

i.e. exactly the structure of a conformal (Casimir) tensor on a circle. This form is imposed by the isotropy of the four extended dimensions plus one compact dimension: it is therefore partly kinematic, and the physical content resides in the *sign* and the *magnitude*.

### 5.2 Sign

[to check] (standard result) A massless *periodic* field in 5D has a negative Casimir energy [10],

$$\rho_{\text{Cas}} = -\frac{\Gamma(5/2) \zeta(5)}{\pi^{5/2} L^5} < 0, \quad (9)$$

which gives  $\rho + p_\sigma < 0$  along the circle: precisely the NEC violation required by (7). The “periodic” nature is that of the Kaluza–Klein fields [6, 7] themselves and of the 5D graviton: no new ingredient (Fig. 2).

### 5.3 Magnitude and self-consistency

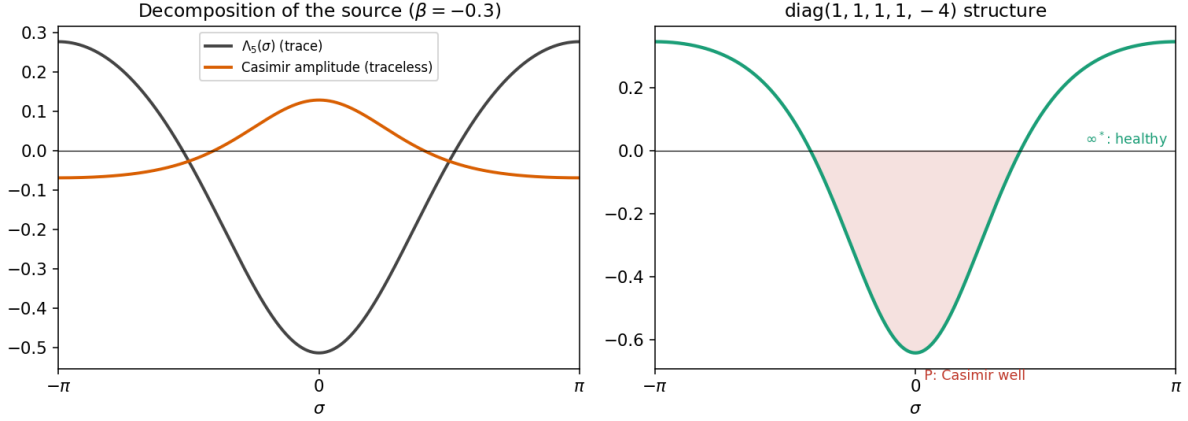
[numerical] The required exotic density at  $P$  is  $\simeq 0.64/(\kappa^2 \ell^2)$ . Equating it to the Casimir density of a circle of radius  $\ell$ ,  $\sim 1/\ell^5$ , imposes

$$\ell \sim 1/M_5 = \ell_5, \quad (10)$$

i.e. a circle of Planckian size — which is tautologically the case, since the pole  $P$  is the Planck point. No hierarchy, no fine-tuning: the scale circle is held closed by its own zero-point energy, exactly at Planck. This cleanly closes the quantum  $\leftrightarrow$  gravity thread of the corpus: the same pole  $P$  that cancels the gravitational potential ( $V = 0$ ) is held closed by the quantum vacuum energy. [conjecture]

## 6 Does vacuum energy predict $\beta$ ? The heat-kernel obstruction

If the warp is held by the Casimir energy, can one go further and *predict*  $\beta$  as the value that minimizes the vacuum energy, rather than fitting it to the CMB? This is the heaviest computation of this work — explicit KK spectrum, Casimir energy regularized by the spectral zeta function, and a test of *dynamical* stabilization of the warp — and we validate each step before drawing a conclusion. We show that the answer is no, for a rigorous reason.



**Figure 2.** [numerical] *Left:* decomposition of the source into a trace part  $\Lambda_5(\sigma)$  and a traceless part (Casimir amplitude  $\frac{3}{5}(\ln a)''$ ). *Right:* the NEC combination  $\rho + p_\sigma$ , of structure  $\text{diag}(1, 1, 1, 1, -4)$ , violated at the Casimir well centered on  $P$  and healthy toward  $\infty^*$ . Sign, structure and magnitude ( $\ell \sim \ell_5$ ) coincide with those of the Casimir energy of periodic fields.

### 6.1 KK spectrum on the warped circle

[numerical] A 5D scalar field separated into modes,  $\Phi = \sum_n \phi_n(x) \psi_n(\sigma)$ , leads to the Sturm–Liouville problem

$$-(a^4 \psi_n')' = \lambda_n a^2 \psi_n, \quad \lambda_n = M_n^2 \ell^2, \quad (11)$$

on the circle, with periodic conditions. The zero mode ( $\lambda_0 = 0$ ) is the 4D graviton; the KK tower *compresses* as  $|\beta|$  grows, the high modes tending to  $(n/L_*)^2$  with the effective size  $L_* = \frac{1}{2\pi} \oint d\sigma/a > 1$  (Fig. 3). The fundamental mode stays pinned at  $M_1 = 1/\ell$  for all  $\beta$ .

| $\beta$ | $M_1$ | $M_2$ | $M_3$ | $M_4$ | $M_5$ | $M_6$ | $L_*$  |
|---------|-------|-------|-------|-------|-------|-------|--------|
| 0       | 1.000 | 2.000 | 3.000 | 4.000 | 5.000 | 6.000 | 1.0000 |
| -0.15   | 1.000 | 1.993 | 2.988 | 3.984 | 4.979 | 5.975 | 1.0041 |
| -0.30   | 1.000 | 1.972 | 2.952 | 3.934 | 4.915 | 5.898 | 1.0176 |
| -0.45   | 1.000 | 1.935 | 2.888 | 3.844 | 4.801 | 5.759 | 1.0427 |

**Table 1.** [numerical] Kaluza–Klein masses  $M_n \ell$  on the warped scale circle (spectral Fourier solver). Validation at  $\beta = 0$ :  $M_n = n$  exactly. The warp compresses the tower as  $|\beta|$  grows (high modes  $\rightarrow (n/L_*)^2$ ), while the fundamental stays pinned at  $M_1 = 1/\ell$ .

### 6.2 Validated machinery, and the nature of the stabilization

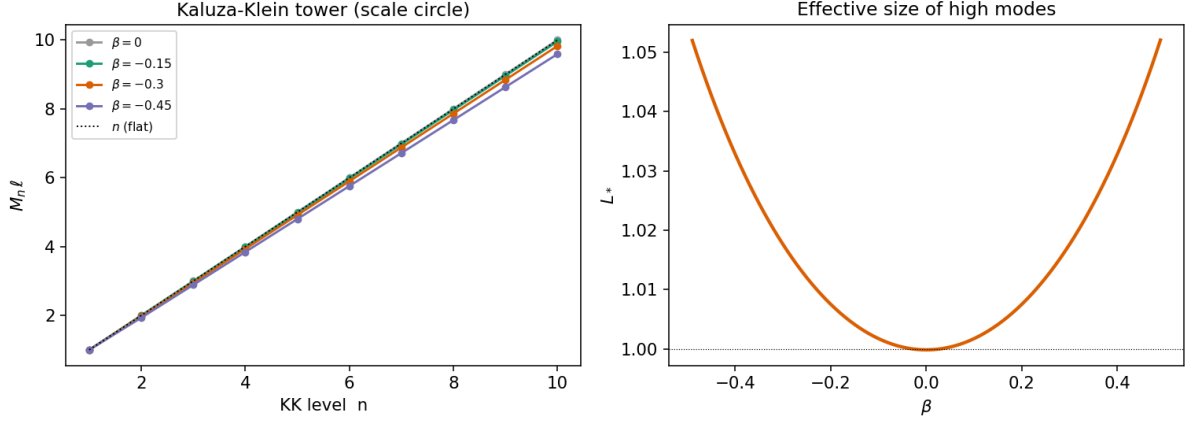
[proof] The total Casimir energy per 4D volume, for a periodic scalar field, is

$$\mathcal{E}(\beta) = \frac{1}{120\pi^2 \ell^5} \zeta_L(-5/2; \beta), \quad \zeta_L(s) = \sum_{n \geq 1} \lambda_n^{-s}, \quad (12)$$

where  $\zeta_L$  is the spectral zeta function of the operator (11). *Validation at  $\beta = 0$ :* since the spectrum there is exactly  $\lambda_n = n^2$  (doubly degenerate, cf. Table 1), one recovers

$$\zeta_L(-5/2; 0) = 2\zeta(-5) = -\frac{1}{126}, \quad (13)$$

— the zeta-regularization machinery is therefore correct in this controlled limit, and the energy there is negative (attractive Casimir, consistent with  $\rho_{\text{Cas}} < 0$  of §5).



**Figure 3.** [numerical] Left: Kaluza–Klein tower  $M_{n\ell}$  on the warped scale circle, for several  $\beta$  (validation at  $\beta = 0$ :  $M_n = n$ ). The warp compresses the high modes. Right: the effective size  $L_*(\beta) = \frac{1}{2\pi} \oint d\sigma/a$ , even in  $\beta$ .

*The nature of the problem.* The lift  $\sigma \rightarrow \sigma + \pi$  exchanges  $\beta \rightarrow -\beta$  while being an isometry of the circle; the energy is therefore *even*,

$$\mathcal{E}(\beta) = \mathcal{E}(-\beta) = \mathcal{E}_0 + \mathcal{E}_2 \beta^2 + \mathcal{E}_4 \beta^4 + \dots \quad (\text{verified: } L_*(-0.3) = L_*(+0.3)). \quad (14)$$

Decisive consequence: the Casimir energy *cannot* fix  $\beta$  at first order. The uniform circle  $\beta = 0$  is an extremum, and  $\beta$  appears only at second order — this is a *moduli stabilization* problem (à la Coleman–Weinberg), where the Casimir contribution  $\mathcal{E}_2$  competes with the (positive) gravitational cost of curving the circle,  $\beta_*$  emerging from the balance at orders  $\beta^2, \beta^4$ . It is precisely this framework that the next subsection defeats: the coefficient  $\mathcal{E}_2$  itself is ill-defined (divergent).

### 6.3 The obstruction: ultraviolet divergence at the order that matters

[proof] [numerical] In Schrödinger form, (11) becomes  $-v'' + Q(\sigma)v = \lambda v$  with the effective potential (closed form)

$$Q(\sigma) = \frac{3}{16} \frac{f'^2}{f} + \frac{3}{4} f''. \quad (15)$$

The Seeley–DeWitt coefficients [12, 13] of this operator control the poles of  $\zeta_L(s)$ :  $\zeta_L(s)$  has a pole at  $s = \frac{1}{2} - k$  with residue  $\propto b_k$ , where

$$b_1 = -\int Q dt, \quad b_2 = \frac{1}{2} \int Q^2 dt, \quad b_3 = \int \left( -\frac{1}{6} Q^3 + \frac{1}{12} \dot{Q}^2 \right) dt. \quad (16)$$

The energy lives at  $s = -5/2$ , i.e.  $k = 3$ . The computation gives (Fig. 4)

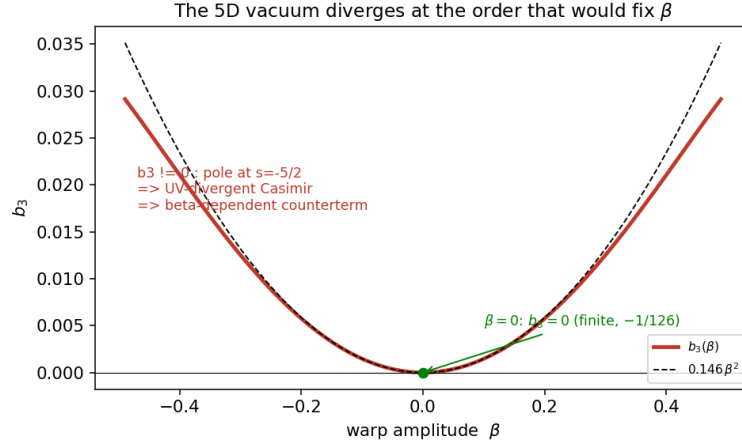
$$b_3(\beta) \simeq 0.146 \beta^2 \neq 0 \quad (\beta \neq 0), \quad b_3(0) = 0 \quad (17)$$

the dominant term coming from  $\int \dot{Q}^2 dt \propto \beta^2$  (a Korteweg–de Vries invariant, robustly non-zero). Chain of consequences, each link verified:

$$b_3 \neq 0 \implies \zeta_L \text{ has a pole at } s = -\frac{5}{2} \implies \mathcal{E} \propto \zeta_L(-\frac{5}{2}) \text{ diverges} \implies \text{a bulk counterterm is required.} \quad (18)$$

### 6.4 Consequence and escape route

[proof] The vacuum energy that would fix  $\beta$  therefore requires a local counterterm *depending on*  $\beta$  (a higher-curvature term in the 5D action). Since its coefficient is free, the renormalized energy carries a scale dependence  $\propto b_3(\beta) \ln \mu$ : the value  $\beta_*$  depends on the renormalization scheme. This behavior is the known one of Casimir energies in warped 5D (Randall–Sundrum type) [9], which renormalize brane tensions and



**Figure 4.** [numerical] The third heat-kernel coefficient  $b_3(\beta)$ , which controls  $\zeta_L(-5/2)$ . Non-zero for  $\beta \neq 0$  (dominant law  $0.146\beta^2$ ), it signals a pole at  $s = -5/2$ : the 5D Casimir energy is ultraviolet-divergent, with a divergence depending on  $\beta$ . At  $\beta = 0$  (flat circle),  $b_3 = 0$  and the energy is finite ( $-1/126$ ).

the bulk cosmological constant. The only escape route would be a *supersymmetric* field content, where  $\sum_F (-1)^F b_3 = 0$  would render the energy finite; but  $\beta_*$  would then remain *content-dependent*. In all cases: no parameter-free prediction. [conjecture]

## 6.5 A point in favor of the $\sigma$ -model

[proof] The original corpus formulation treats  $\sigma$  as an internal target: there is then *no* bulk circle on which fields wind, hence no analogous Casimir divergence. On this precise point, the  $\sigma$ -model is better behaved than its 5D promotion. The dimensional embedding illuminates the *origin* of the action, but does not provide better control of  $\beta$ .

## 7 Discussion

### 7.1 What the 5D embedding brings, and what it does not

The assessment is mixed and instructive. The embedding *regenerates* the corpus structure: the non-minimal coupling  $f = a^2$ , its trigonometric form, the  $V$ - $f$  locking, and the F2 health criterion become consequences of the warped 5D geometry rather than choices. It also provides a *natural origin* for the required exotic source: vacuum Casimir energy, with correct sign, structure and Planckian magnitude. But it *does not improve* predictive power: the coupling  $\beta$  cannot be predicted from the vacuum energy, whose relevant part is UV-divergent. The embedding is thus to be read as a structural *origin story*, not as a more fundamental framework replacing the  $\sigma$ -model.

### 7.2 Positioning in the literature and related work

None of the physical bricks mobilized here is proper to this work: each has an identifiable lineage, which we first survey, before isolating what is, properly, a matter of *assembly*.

- *Scale as a variable.* Promoting resolution/scale to the rank of a coordinate is the guiding idea of Nottale’s scale relativity [17]; on the holographic side, the identification “radial coordinate = energy scale / renormalization direction” (1) is an established fact of AdS/CFT [8, 18].
- *Inflation.* Our inflationary sector (periodic non-minimal coupling  $f = 1 + \beta \cos$ ,  $V \propto 1 - \cos$ ) coincides with the periodic non-minimal-coupling natural inflation of Ferreira–Notari–Simeon [15] and Simeon [16], which brings “bare” natural inflation [14] — today disfavored by the data — back within the  $(n_s, r)$  contours at the GUT scale.
- *c-function / directed flow.* The link “null energy condition  $\Rightarrow$  a monotone function along the flow” goes back to the Zamolodchikov c-theorem [19] and its holographic version [20]; our cosmological

c-function is an FRW adaptation of it.

- *Bulk Casimir*. The Casimir energy of a compact dimension, regularized over the KK tower, and its tendency to stabilize or destabilize the radius, are classical since Appelquist–Chodos [11].
- *Scale duality*. The UV  $\leftrightarrow$  IR inversion with a minimal length (“zero-point length”,  $ds \rightarrow \ell_P^2/ds$ ) of Padmanabhan [21], a cousin of string T-duality, prefigures our involution  $\mathcal{D} : \sigma \rightarrow -\sigma$  (1).
- *Relational clock*. The bridge whereby gravity modulates the rate of an internal clock is the Page–Wootters formalism [22] (sector developed in Units 1 and 4–5).
- *Closure*. That a regular black hole requires an energy-condition violation (de Sitter core) is established since Bardeen [23] and Hayward [24].

#### What is genuinely ours — new as assembly, not as physics

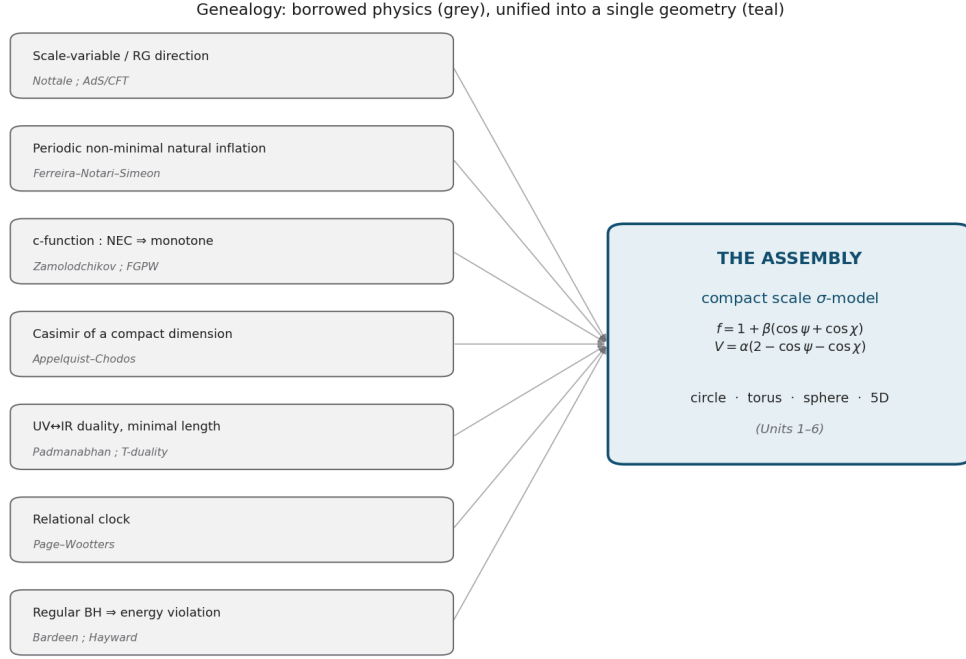
None of the ingredients above is new in isolation, and we claim *no new observable*: the predicted  $(n_s, r)$  are degenerate with non-minimal natural inflation [15, 16]. What is proper to the corpus comes down to three points.

1. **The unifying move.** Compactifying *both* scale axes — spatial and temporal — as internal targets of a single scalar–tensor  $\sigma$ -model ( $f = 1 + \beta(\cos \psi + \cos \chi)$ ,  $V = \alpha(2 - \cos \psi - \cos \chi)$ ), then realizing the compact object as a circle/torus (Units 1–4) [1, 2, 3, 4], a sphere (Unit 5) [5], or a warped fifth dimension (this Unit). We have not found this precise construction elsewhere: it is *in the spirit* of Nottale [17] and of holographic RG [8, 18], but its realization as a scale  $\sigma$ -model is proper to the corpus.
2. **Over-determination.** A single coupling  $\beta$ , fixed once by the CMB, must *simultaneously* satisfy several sectors (inflation, static closure, false-vacuum metastability, clock freeze, core stability, RG flow), through crossed internal identities — the same threshold  $\beta_c$  pivoting two sectors at once. This is the methodological signature of the programme, and its possibly fortuitous character is explicitly left [conjecture] / [open].
3. **The model-internal exact forms.** A closed-form Hawking–Moss action vanishing at the threshold, the spectral validation  $\zeta_L(-5/2; 0) = -1/126$ , the heat-kernel coefficient  $b_3(\beta) \simeq 0.146\beta^2$ , the existence factor  $\varepsilon = E(k)/K(k)$ : exact results, but *internal* to the framework.

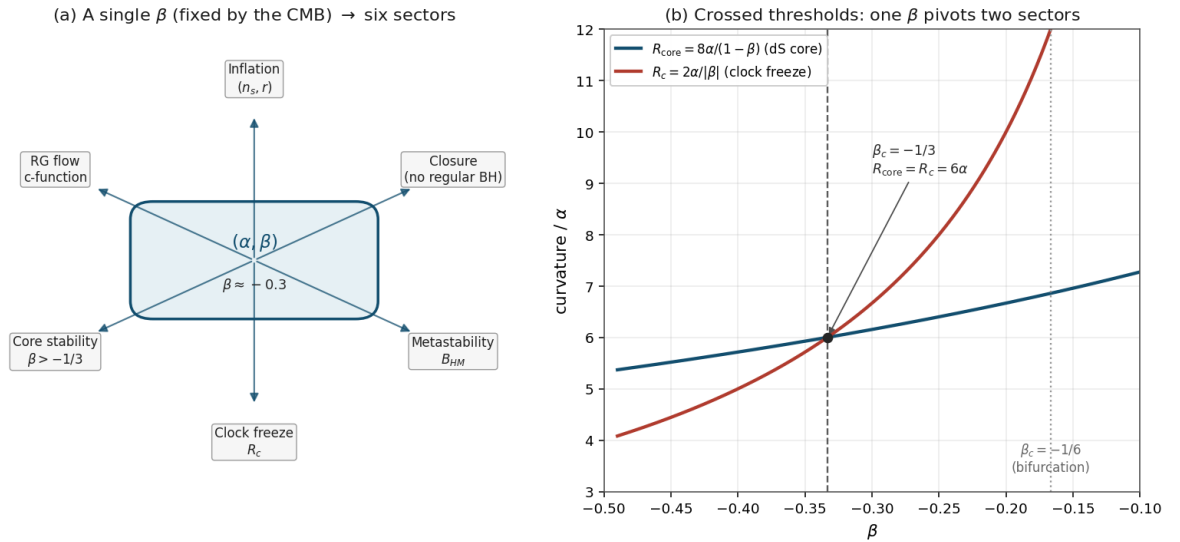
In sum, the contribution is not a distinctive prediction but a *grammar*: a single compact geometry that reorganizes disparate phenomena under a handful of numbers, each statement carrying its epistemic status. Anything distinctive lives at GUT/Planck curvature (screened) or remains structurally real but silent in the currently accessible spectrum.

### 7.3 Mapping the assembly

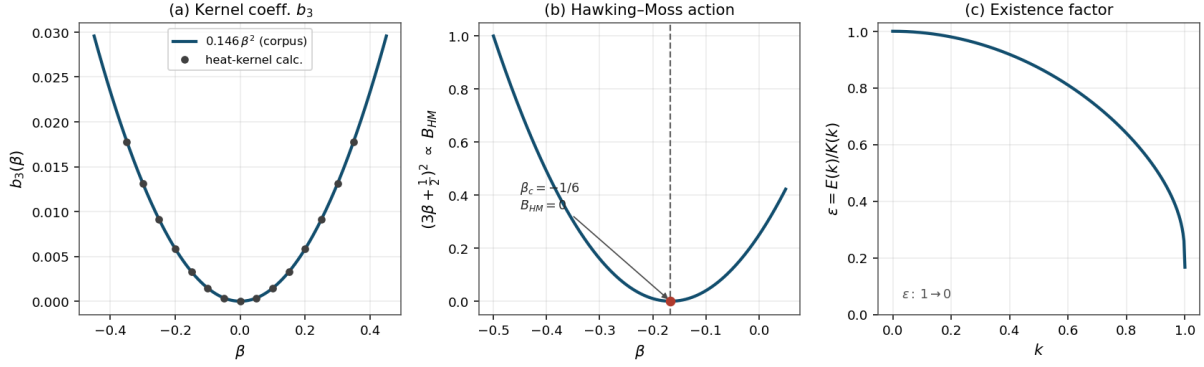
These three traits are read directly off Figures 5–7. The genealogy (Fig. 5) separates the borrowed bricks — each of identifiable lineage — from their systematization, which is the proper object. The over-determination (Fig. 6) shows a single  $(\alpha, \beta)$  feeding six sectors, and the exact interlocking of thresholds ( $\beta_c = -1/3$  where  $R_{\text{core}} = R_c = 6\alpha$ ; bifurcation at  $-1/6$ ). Finally the model-internal closed forms (Fig. 7) —  $\zeta_L(-5/2; 0) = -1/126$ ,  $b_3 \simeq 0.146\beta^2$ ,  $B_{\text{HM}} \propto (3\beta + \frac{1}{2})^2$  vanishing at  $-1/6$ ,  $\varepsilon = E/K$  — are exact and reproducible, but proper to the framework. All three are regenerated by the script `U6_appartenance.py` (§ Reproducibility).



**Figure 5.** *Genealogy of the assembly.* Each physical brick (grey) has an identifiable lineage in the literature; the corpus unifies them into a single compact geometry (teal). The proper contribution is not a brick, but their systematization.



**Figure 6.** *Over-determination.* (a) A single  $(\alpha, \beta)$ , fixed by the CMB, feeds six sectors. (b) Interlocking of thresholds:  $R_{core}(\beta)$  and  $R_c(\beta)$  cross exactly at  $\beta_c = -1/3$  ( $R = 6\alpha$ ); the false-vacuum bifurcation is at  $\beta_c = -1/6$ . A single parameter governs two distinct sectors.



**Figure 7.** Model-internal exact forms. **(a)**  $b_3(\beta)$ : the corpus law  $0.146\beta^2$ , with the points recomputed from the heat kernel. **(b)**  $B_{HM} \propto (3\beta + \frac{1}{2})^2$ , vanishing at  $\beta = -1/6$ . **(c)**  $\varepsilon = E(k)/K(k)$ , decreasing from 1 to 0. Exact, reproducible results, but proper to the framework.

## 8 Epistemic status and limitations

| Content                                                                                                                                                                                                                                                                                              | Status              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| KK reduction $\Rightarrow f = a^2$ ; trigonometric form forced by compactness; $V$ - $f$ locking giving $V = \alpha(1 - \cos \sigma)$ ; F2 criterion re-derived; 5D Einstein tensor and NEC combination (7); AdS <sub>5</sub> identification for $a = e^{\sigma/\ell}$ .                             | [proof] [numerical] |
| Geometric structure: AdS <sub>5</sub> as a dilatation isometry ( $R_5 = -20/\ell^2$ , Killing $\partial_\sigma$ , scale momentum); duality involution $\mathcal{D}: \sigma \rightarrow -\sigma$ (T-duality, fixed points $P, \infty^*$ ); compactness breaks dilatation and generates the hierarchy. | [proof]             |
| Geometric origin of the Unit 1 clock (KK tower $\rightarrow \tau_0, \varepsilon$ ): scale coherence, but not derived in this work.                                                                                                                                                                   | [conjecture] [open] |
| KK spectrum on the warped circle; compression and $L_*$ ; validation $\zeta_L(-5/2; 0) = -1/126$ ; $\mathcal{E}(\beta) = \mathcal{E}(-\beta)$ ; coefficients $b_1, b_2, b_3$ ; $b_3(\beta) \simeq 0.146\beta^2$ ; UV divergence at $s = -5/2$ .                                                      | [proof] [numerical] |
| Exotic source identified with Casimir energy (sign, structure, self-consistent Planckian magnitude).                                                                                                                                                                                                 | [conjecture]        |
| $\beta_*$ not predicted parameter-free: scheme dependence (counterterm $\propto b_3$ ), supersymmetric escape route but content-dependent.                                                                                                                                                           | [conjecture] [open] |
| Multi-field resolution off the diagonal; exact behavior near the poles; degeneracy at the confluence point $\infty^*$ ; explicit computation of the 5D counterterms; the ghost question beyond the signature choice.                                                                                 | [open]              |

The distinction of statuses is not an ornament but a constitutive part of the method: it guarantees that this work can be reused without any statement exceeding what is actually established.

## 9 Conclusion

Promoting the scale axis to the rank of a compactified spacetime fifth dimension — the inverse of the corpus choice — lands first on marked-out geometric ground: the warped metric is AdS<sub>5</sub>, the scale is its renormalization direction,  $\partial_\sigma$  is its dilatation isometry, and the involution  $\sigma \rightarrow -\sigma$  is a scale mirror (cousin of T-duality) leaving  $P$  and  $\infty^*$  fixed. Closing this scale line into a loop breaks the exact dilatation — which, far from being a defect, *generates the hierarchy* of scales.

The embedding is then *generative*: the Kaluza–Klein reduction reproduces the non-minimal coupling  $f = a^2$ , its trigonometric form imposed by compactness, the  $V$ - $f$  locking giving  $V = \alpha(1 - \cos \sigma)$ , and F2 health. It even offers a *bridge* toward Unit 1: the discreteness of the KK tower is the geometric candidate for the origin of the intermittent clock  $(\tau_0, \varepsilon)$  — coherent in order of magnitude, but *not derived* here, and we record it as such.

Two costs, in all honesty. On the one hand, the promotion *reopens* the ghost and closed-timelike-curve question that F2 closed by postulate; we close it again, but only by keeping the scale on the *space* side. On the other hand, closing the circle requires a null-energy-condition violation at the Planck pole — a cost that naturally identifies with vacuum Casimir energy, with correct sign, structure and Planckian magnitude. But the embedding *does not improve* predictive power: the vacuum energy does not fix  $\beta$ , its relevant part being UV-divergent ( $b_3 \neq 0$ ), so that  $\beta_*$  depends on the scheme; on this point, the original  $\sigma$ -model is even better behaved.

The 5D embedding is therefore an illuminating *structural origin* — it recovers and grounds the form of the framework — and not a more fundamental replacement. This is the price at which, with the honest labeling of each statement, a geometric intuition can be evaluated for what it is worth: here, a fine origin and a promising bridge, but not a prediction.

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## Reproducibility

The numerical results and symbolic verifications are reproduced by two standalone scripts (NumPy/SymPy/SciPy/Matplotlib), with sections coded to be proper to this document:

- [W0] Geometric structure:  $\text{AdS}_5$  for  $a = e^{\sigma/\ell}$  ( $R_5 = -20/\ell^2$  constant); dilatation isometry (Killing  $\partial_\sigma$ ); duality involution  $\mathcal{D} : \sigma \rightarrow -\sigma$  and its fixed points  $P, \infty^*$ .
- [W1] KK reduction: 5D Einstein tensor (2), induced coupling  $f = a^2$ ,  $V$ - $f$  locking (6), absence of a second harmonic (brane-tension test); F2 criterion (sign of the kinetic term according to the signature of  $\sigma$ ).
- [W2] NEC combination (7); localization of the violation at  $P$ .
- [W3] Decomposition of the source: trace part  $\Lambda_5(\sigma)$  and traceless part  $\propto \text{diag}(1, 1, 1, 1, -4)$  (8); sign of the periodic Casimir (9); self-consistency  $\ell \sim \ell_5$  (10).
- [W4] Warped KK spectrum (11) (spectral Fourier solver); compression and  $L_*$ ; symmetry  $\beta \rightarrow -\beta$ ; validation  $\zeta_L(-5/2; 0) = 2\zeta(-5) = -1/126$ .
- [W5] Schrödinger form (15); heat-kernel coefficients (16);  $b_3(\beta) \simeq 0.146\beta^2$  (17); pole structure and UV divergence (18).
- [W6] Mapping the assembly (Fig. 5–7): recomputed cross-checks —  $\zeta_L(-5/2; 0) = 2\zeta(-5) = -1/126$ ;  $b_3(\beta)$  by integration of the heat kernel  $\rightarrow 0.146\beta^2$  (limit  $3\pi/64$ );  $B_{\text{HM}} \propto (3\beta + \frac{1}{2})^2$  vanishing at  $\beta = -1/6$ ; interlocking  $R_{\text{core}} = R_c$  at  $\beta = -1/3$  ( $R = 6\alpha$ );  $\varepsilon = E/K$ .

Files: `U6_5D_reproducible.py` (computations), `U6_figures.py` (Fig. 1–4) and `U6_appartenance.py` (cartography, Fig. 5–7). Convention:  $\ell = M_5^{-1} = 1$  unless stated otherwise; the prefix `U6_5D_` and the codes [W·] are specific to this document.

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